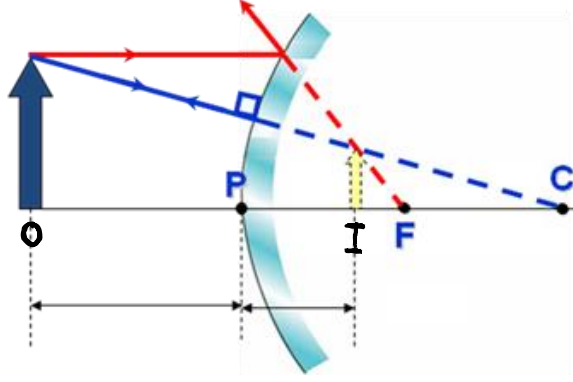


NO.	ANSWER SCHEME	MARKS
1(a) (i)	$A = 0.5 \text{ m}$	JU1
(ii)	$\omega = \frac{\pi}{5} \text{ rad s}^{-1} @ \omega = 0.628 \text{ rad s}^{-1}$	JU1
(iii)	$k = 5\pi = 15.71 \text{ m}^{-1}$	JU1
(iv)	$\omega = 2\pi f$ $\frac{\pi}{5} = 2\pi f$ $f = 0.1 \text{ Hz}$	G1 JU1
(v)	$k = \frac{2\pi}{\lambda}$ $5\pi = \frac{2\pi}{\lambda}$ $\lambda = 0.4 \text{ m}$	G1 J1
(vi)	$v = \lambda f$ $= (0.1)(0.4)$ $= 0.04 \text{ m s}^{-1}$	G1 JU1
(b) (i)	<i>mass per unit length</i> , $\mu = \frac{m}{l}$ $= \frac{220 \times 10^{-3}}{1.6}$ $= 0.1375 \text{ kgm}^{-1}$	K1 G1 JU1
(ii)	<i>velocity</i> , $v = \sqrt{\frac{T}{\mu}}$ $= \sqrt{\frac{120}{0.1375}}$ $= 29.5 \text{ ms}^{-1}$	G1 JU1
(iii)	$v = f\lambda$ $29.5 = 30\lambda$ <i>wavelength</i> , $\lambda = 0.98 \text{ m}$	G1 JU1
	TOTAL	16

NO.	ANSWER SCHEME	MARKS
2(a) (i)	$V = IR$ $10 = (0.80)R$ $R = 12.5 \Omega$	G1 JU1
(ii)	$\rho = \frac{RA}{l}$ $= \frac{(12.5)\pi(0.5 \times 10^{-3})^2}{3.5}$ $= 2.805 \times 10^{-6} \Omega m$	G1 JU1
(b)	$I = \frac{Q}{t}$ $Q = ne$ or $I = \frac{ne}{t}$ $n = \frac{It}{e}$ $= \frac{(5.0)(3600)}{1.6 \times 10^{-19}}$ $= 1.13 \times 10^{23} \text{ electrons}$	K1 G1 JU1
(c)	$R = \frac{\rho l}{A}$ $(3.5) = \frac{(2.5 \times 10^{-5})l}{\frac{\pi(1.5 \times 10^{-3})^2}{4}}$ $l = 0.247 \text{ m}$	G1 JU1
(d)	$R_{12} = R_1 + R_2$ $= 5 + 5$ $R_{12} = 10 \Omega$ $\frac{1}{R_{eff}} = \frac{1}{R_{12}} + \frac{1}{R_3} + \frac{1}{R_4}$ $= \frac{1}{10} + \frac{1}{2} + \frac{1}{4}$ $R_{eff} = 1.18 \Omega$	G1 JU1
	TOTAL	12

NO.	ANSWER SCHEME	MARKS
3. (a)(i)	$C_{12} = C_1 + C_2$ $C_{12} = 6 \times 10^{-6} + 5 \times 10^{-6}$ $C_{12} = 11 \times 10^{-6} \text{ F}$ $\frac{1}{C_{eff}} = \frac{1}{C_{12}} + \frac{1}{C_3}$ $\frac{1}{C_{eff}} = \frac{1}{11 \times 10^{-6}} + \frac{1}{3 \times 10^{-6}}$ $C_{eff} = 2.357 \times 10^{-6} \text{ F}$	 G1 G1 JU1
(ii)	$Q_1 = C_1 V_1$ $(50 \times 10^{-6}) = (6 \times 10^{-6}) V$ $V_1 = 8.33 \text{ V}$	 G1 JU1
3(b) (i)	$Q_o = CV$ $= (6 \times 10^{-6})(12)$ $= 7.2 \times 10^{-5} \text{ C}$	 G1 JU1
(ii)	$\tau = RC$ $= (35)(6 \times 10^{-6})$ $= 2.1 \times 10^{-4} \text{ s}$	 GJU1
(iii)	$Q = Q_o \left(1 - e^{-\frac{t}{RC}} \right)$ $= (7.2 \times 10^{-5}) \left(1 - e^{-\frac{4.2 \times 10^{-4}}{(2.1 \times 10^{-4})}} \right)$ $= 6.23 \times 10^{-5} \text{ C}$	 G1 JU1
	TOTAL	10

No	Answer Scheme	Marks
4. (a) (i)	$B = \frac{\mu_o I}{2\pi R}$ $= \frac{(4\pi \times 10^{-7})(3.5)}{2\pi(2.0 \times 10^{-2})}$ $= 3.5 \times 10^{-5} T$	G1 JU1
(a) (ii)	$B = \frac{\mu_o NI}{L}$ $L = \frac{(4\pi \times 10^{-7})(30)(20 \times 10^{-3})}{1.5 \times 10^{-6}}$ $= 0.5 m$	G1 JU1
(b) (i)	$\phi = BA \cos \theta$ $= (2.5)(6 \times 10^{-4}) \cos 0$ $= 1.5 \times 10^{-3} Wb$	G1 JU1
(b) (ii)	$\phi = BA \cos \theta$ $= (2.5)(6 \times 10^{-4}) \cos 30$ $= 1.3 \times 10^{-3} Wb$ Change in magnetic flux = $(1.3 \times 10^{-3} - 1.5 \times 10^{-3})$ $= -0.2 \times 10^{-3} Wb \quad @ \quad -2.0 \times 10^{-4} Wb$	GJ1 K1 JU1
(c) (i)	$\Delta \phi = \phi_f - \phi_i = (B_f - B_i)A \cos \theta$ $= (0.8 - 0.650)\pi(3 \times 10^{-2})^2 \cos 30 = 3.67 \times 10^{-4} Wb$ $\varepsilon = -\frac{d\phi}{dt}$ $= -\frac{3.67 \times 10^{-4}}{0.5}$ $= -7.35 \times 10^{-4} V$	GJ1 G1 JU1
(c) (ii)	$\varepsilon = IR \Rightarrow I = \frac{\varepsilon}{R}$ $= \frac{7.35 \times 10^{-4}}{8}$ $= 9.19 \times 10^{-5} A$	G1 JU1
	TOTAL	14

No	Solution	Marks
5 (a) i)		D-2M (2 rays -1M Label O & I - 1M)
5 (a) ii)	$f = -15.0 \text{ cm}$ (convex mirror), $v = -8.0 \text{ cm}$ (image is virtual) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{-15.0} = \frac{1}{u} + \frac{1}{-8.0}$ $u = 17.14 \text{ cm}$	G1 JU1
5 (b) (i)	$m = -\frac{v}{u}$ $v = -3u$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{10.0} = \frac{1}{u} + \frac{1}{-3u}$ $u = 6.67 \text{ cm} \quad \text{or} \quad 6.67 \times 10^{-2} \text{ m}$	G1 JU1
(ii)	$m = -\frac{v}{u}$ $v = -3(6.67)$ $= -20 \text{ cm} \quad \text{or} \quad -0.2 \text{ m}$ $\text{- at the back of the mirror}$	G1 JU1 J1
5 (c) i)	$u = 25 \text{ cm}$, $f = 8 \text{ cm}$, $h_o = 5 \text{ cm}$ $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\frac{1}{8} = \frac{1}{25} + \frac{1}{v}$ $v = +11.76 \text{ cm}$ $\text{(opposite side of the object or behind the lens)}$	G1 JU1 J1

ii)	$m = \frac{h_i}{h_o} = -\frac{v}{u}$ Image form inverted → $-\frac{h_i}{5} = -\frac{11.76}{25}$ $h_i = 2.35 \text{ cm}$	G1 JU1
	TOTAL	14

NO	ANSWER SCHEME	MARKS
6 (a) (i)	third maximum, $m = 3$ $y_m = \frac{m\lambda D}{d}$ $= \frac{(3)(500 \times 10^{-9})(0.5)}{0.25 \times 10^{-3}}$ $= 3.0 \times 10^{-3} \text{ m}$	K1 G1 JU1
(ii)	second minimum, $m = 1$ $y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$ $= \frac{\left(1 + \frac{1}{2}\right)(500 \times 10^{-9})(0.5)}{0.25 \times 10^{-3}}$ $= 1.5 \times 10^{-3} \text{ m}$	K1 G1 JU1
(b)(i)	Tenth minimum, $m = 9$ $x_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$ $d = \frac{\left(9 + \frac{1}{2}\right)(600 \times 10^{-9})(1.5)}{8.25 \times 10^{-3}}$ $= 1.04 \times 10^{-3} \text{ m}$	K1 G1 JU1

(ii)	$\Delta y = \frac{\lambda D}{d}$ $= \frac{(600 \times 10^{-9})(1.5)}{1.04 \times 10^{-3}}$ $= 8.65 \times 10^{-4} m$	G1 JU1
(iii)	$\Delta y = \frac{\lambda D}{d}$ $\Delta y \propto \lambda$ λ decrease, Δy also decrease the fringes separation get closer to each other.	K1 K1 J1
	TOTAL	14